Vector Analysis

Lab_03: Vector Digitization & Spatial Queries

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5’ Recap

What did GCPs do?

GCS vs PCS

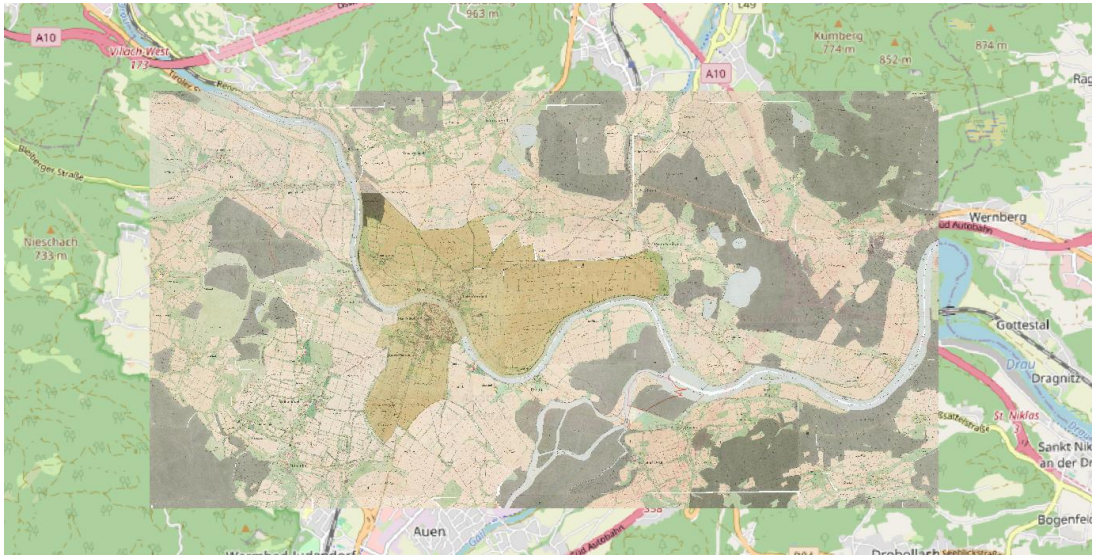
Georeferencing

Equal Earth Projection

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Old maps, modern data

Unless you are going to use your georeferenced map as a simple background image, the next natural step is to digitize elements from it such as buildings, lakes, roads and more



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Change detection example

c. 1826



Silbersee area

Modern map, screenshot from 2024



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Benefits of digitizing old maps

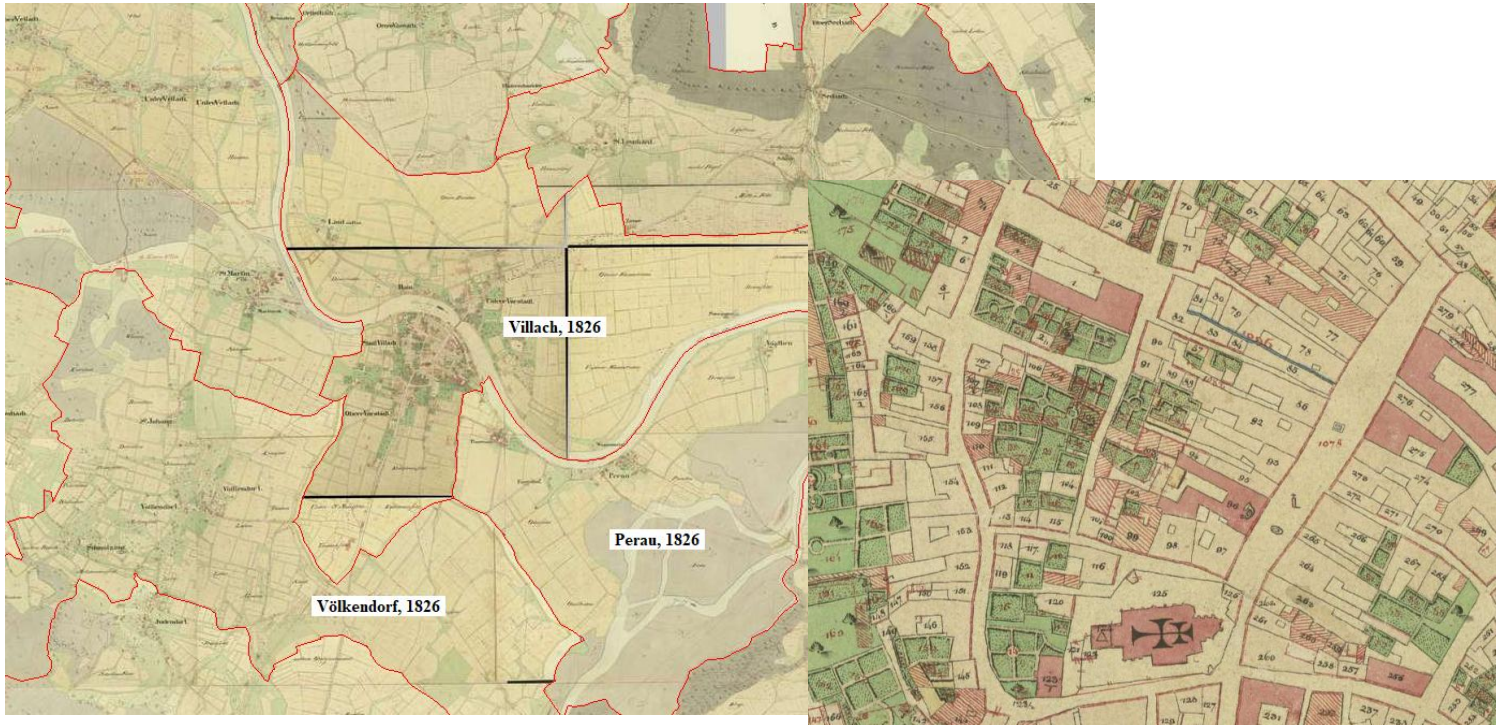
1. Explore local history
2. Analyse mapped features
3. Compare past and present
4. Share reusable data



@RecordsUsa

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Mapire - Cadastral Survey of the Austrian Empire (Franciscan Cadastre)



@ArcanumMaps

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Digitization

What is digitization?

"It is the process of converting the geographic features on a map (analogue, scanned) into digital x, y coordinates, or spatial data". (ESRI GIS Dictionary)

Points for single locations

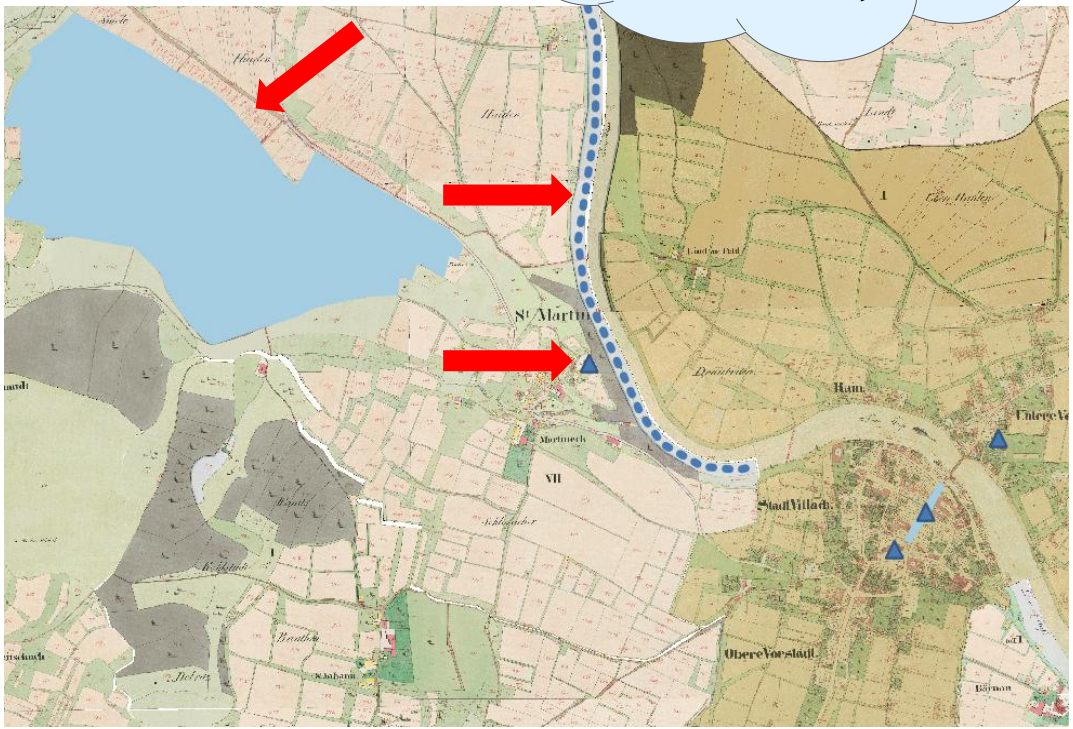
"Where is St. Jakob Church"?

Lines for features with length

"What is the path of the Drau River"?

Polygons for areas

"What is the boundary of this lake"?

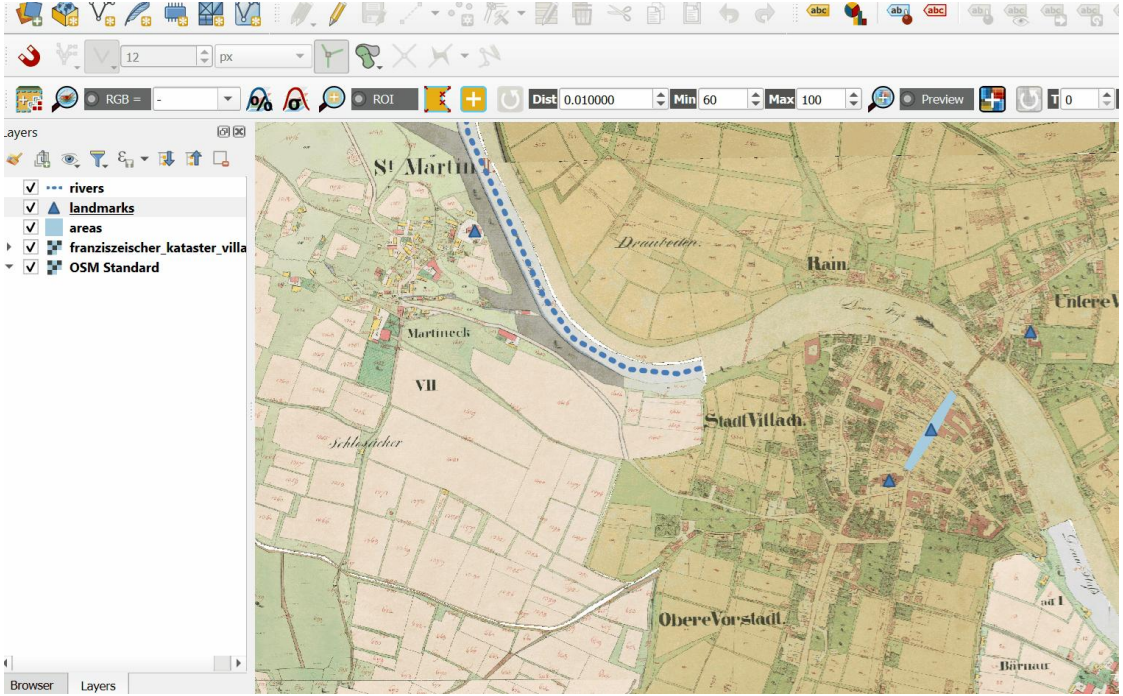


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Importance of attributes in digitizing

Attribute table
Add fields:
-name
-type

Attributes describe features and help us filter them. Geometry allows us to select features by location!



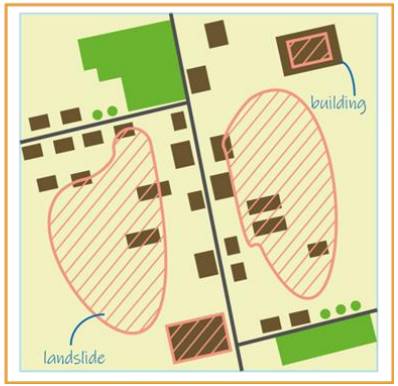
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Spatial queries

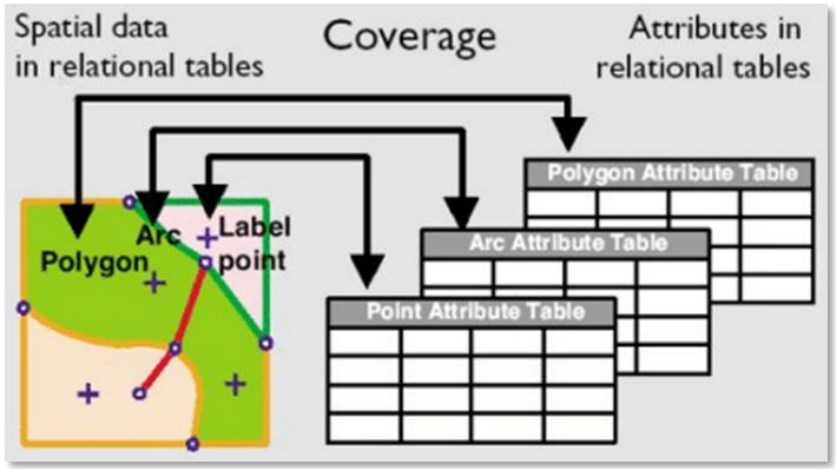
ATTRIBUTE QUERY
(select by attribute)

ID	owner	street	no.	area
1	Jones	Lisson St.	4	96,25
2	Smith	Bell St.	1	112,37
3	Smith	Bell St.	5	147,76
4	Williams	Homer St.	2	128,91
5	Evans	Lisson St.	8	64,28
6	Johnson	Lisson St.	6	281,42
7	Black	Homer St.	10	121,73
8	Evans	Bell St.	7	210,58
9	Thomas	Homer St.	6	88,25
10	Smith	Lisson St.	2	114,87
11	Taylor	York St.	3	73,14

SPATIAL QUERY
(select by location)



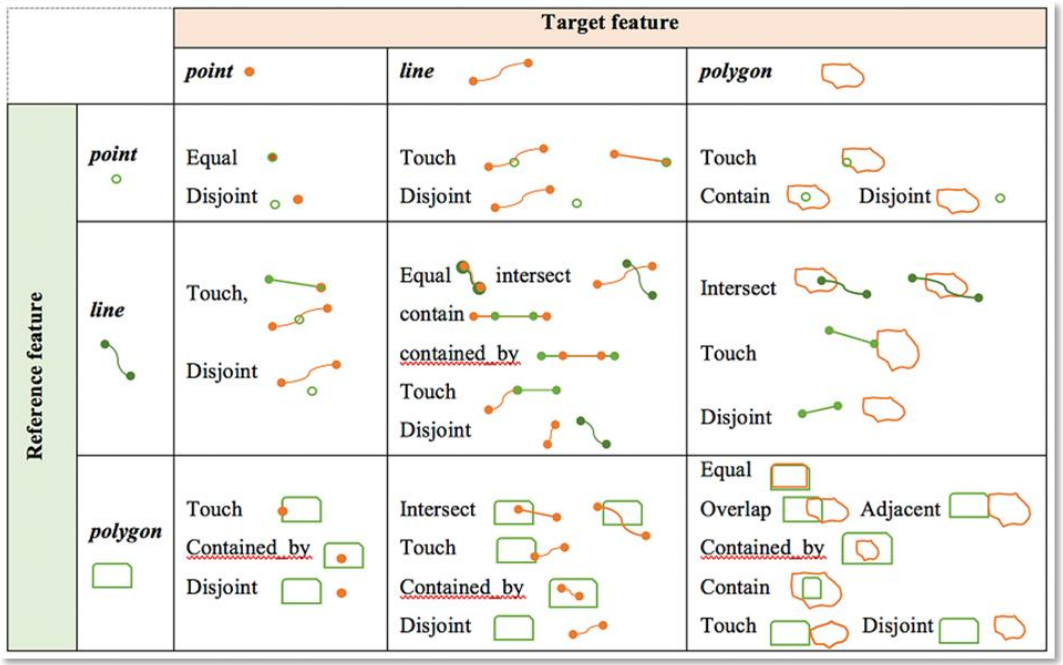
In GIS we use spatial queries to ask location-based questions!



@Saylor Academy

Spatial query by location

- **Intersect**: select features that touch, cross, or overlap
- **Contain**: select areas containing landmarks
- **Are within distance of**: select features close to others
- **Touch**: select features whose boundaries touch



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Buffer

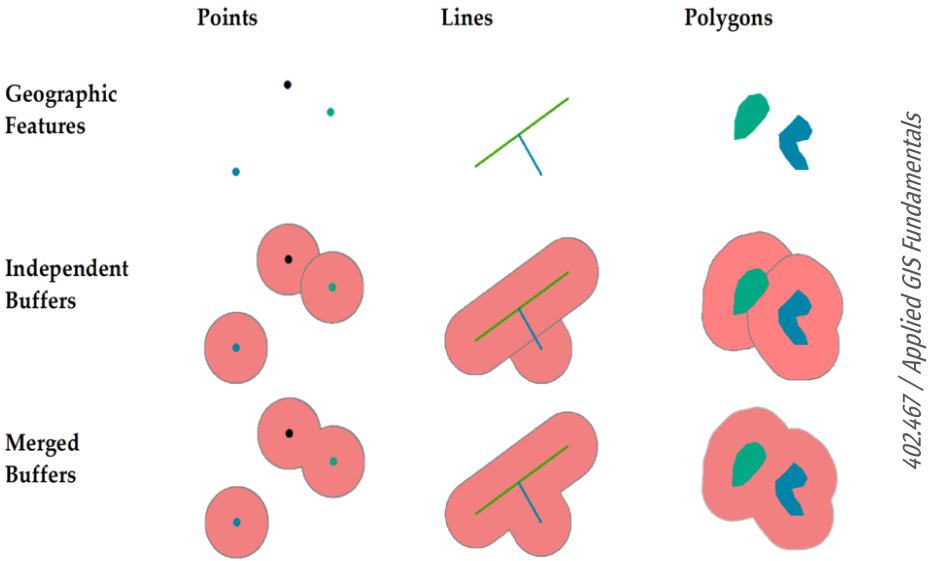
Creates a polygon (a buffer zone) **around** input **features** (points, lines, or polygons) **at a specified distance**

➡️ What's within X distance of...?

Real-world Examples

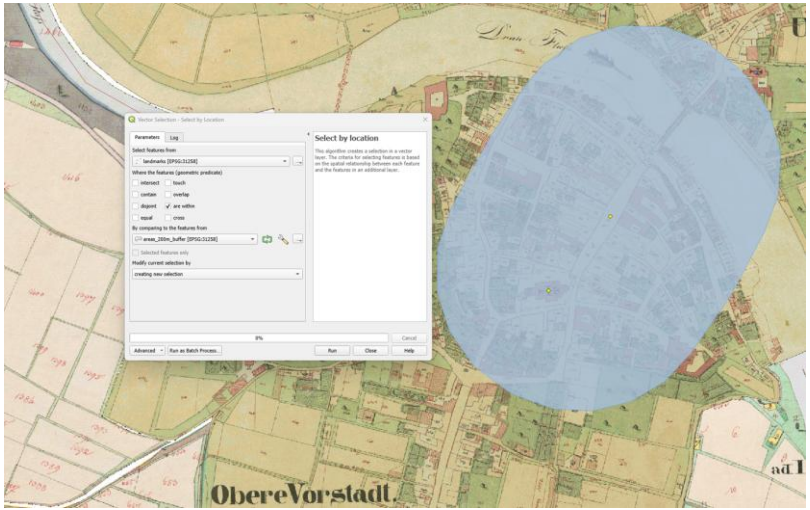
- Protecting rivers (no building within 100m)
- Identifying areas to investigate for potential noise impacts (within 500m of a highway)

The output layer of a buffer operation is always a polygon



Examples

Select by location



400 m buffer around the lake



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Related readings



- Király G, Walz U, Podobnikar T, Czimber K, Neubert M, Kokalj Ž. Georeferencing of historical maps – methods and experiences. In: Csaplovics E, Wagenknecht S, Seiler U, editors. *Spatial Information Systems for Transnational Environmental Management of Protected Areas and Regions in the Central European Space*. Berlin: Rhombos-Verlag; 2008. p. 53-63, available at https://www.researchgate.net/publication/256307982_Georeferencing_of_historical_maps_--_methods_and_experiences
- Pope R-No-A-Rangi, Frean M. Georeferencing Historical Maps at Scale. In: Sila-Nowicka K, Moore A, O'Sullivan D, Adams B, Gahegan M, editors. Proceedings of the 13th International Conference on Geographic Information Science (GIScience 2025); 2025 Aug 26-29; Christchurch, New Zealand. Leibniz Int Proc Informatics (LIPIcs), vol 346:11:1-11:11. DOI: 10.4230/LIPIcs.GIScience.2025.11, available at: https://drops.dagstuhl.de/storage/00lipics/lipics-vol346-giscience2025/LIPIcs.GIScience.2025.11/LIPIcs.GIScience.2025.11.pdf
- Novak A, Ostash V. Digitizing Historical Maps and their presentation in Online Map Collections. E-Perimetron. 2022;17(1):33-44, available at: https://www.e-perimetron.org/Vol_17_1/Novak_Ostash.pdf